Automating Employee Data

1. PROJECT OVERVIEW

1.1 Project Overview

The *Automating Employee Data* project focuses on simplifying and automating the process of managing employee records within an organization using ServiceNow. The system eliminates manual data entry and ensures accurate, consistent, and centralized data management.

1.2 Purpose

The purpose of this project is to:

- Reduce manual effort in employee data management
- Improve data accuracy and consistency
- Automate onboarding and updates
- Provide real-time access to employee details

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2. IDEATION PHASE

2.1 Problem Statement

Manual employee data management leads to errors, duplication, and inefficiency. Organizations need an automated system to manage employee information accurately and efficiently.

2.2 Empathy Map Canvas

User: HR/Admin

- **Says:** “Managing employee data is difficult”
- **Thinks:** Need automation
- **Does:** Manual entry and updates
- **Feels:** Stressed and frustrated

Pain Points:

- Time-consuming

- Error-prone
- No centralized system

Gains:

- Automation
- Accuracy
- Easy access

2.3 Brainstorming

Ideas Generated:

- Excel-based tracking
- Import Sets
- Transform Maps
- Workflow automation
- Dot walking

Final Idea Selected:
Automating Employee Data using ServiceNow

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- Data preparation → Import → Transform → Store → Display → Maintain

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3.2 Solution Requirement

Functional Requirements:

- Upload employee data
- Import into staging table
- Transform into user table

- Display details in incident form

Non-Functional Requirements:

- Accuracy
- Performance
- Security
- Scalability

3.3 Data Flow Diagram

Input: CSV/Excel file
Process: Import Set + Transform Map
Storage: sys_user table
Output: Incident form display

3.4 Technology Stack

- Platform: ServiceNow
- Language: JavaScript (for scripting)
- Database: ServiceNow tables
- Tools: Import Sets, Transform Maps

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4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution directly addresses manual data issues by automating data handling using ServiceNow tools.

4.2 Proposed Solution

- Upload employee data
- Automate processing

- Store in centralized database
- Display dynamically using dot walking

4.3 Solution Architecture

Flow:
User → Import Set → Staging Table → Transform Map → User Table → Incident Form

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Task	Duration
Ideation	Problem analysis	1 week
Design	Architecture & planning	1 week
Implementation	Import & mapping	2 weeks
Testing	Validation	1 week

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Tested data import speed
- Checked accuracy of records
- Verified system performance

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7. RESULTS

7.1 Output Screenshots

(Add screenshots of:)

- Import Set

- Transform Map
- User Table
- Incident Form

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work
- Improves accuracy
- Saves time
- Centralized system

Disadvantages

- Initial setup complexity
- Requires ServiceNow knowledge

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9. CONCLUSION

The project successfully automates employee data management, improving efficiency, accuracy, and reducing manual effort.

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10. FUTURE SCOPE

- Integration with other systems
- Advanced analytics
- AI-based automation
- Mobile access

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11. APPENDIX

Dataset photo

Name	Email	Departmei	Manager	Manager Email
Arun Kuma	arun.kuma	IT	Suresh Bal	suresh.babu@example.com
Priya Sharr	priya.sharr	HR	Neha Vern	neha.verma@example.com

GitHub & Project Demo Link

https://github.com/Ashwath54/automating-employees-data

https://drive.google.com/file/d/1P3Rc7D2_nei4qfqwfDeOj1hi0HRdrO48/view?usp=drivesdk